

Continuous linear maps bounded by a seminorm

`MathlibAnnex.EquivalentSeminorm.IsContraction`

def

Defines the pointwise contraction condition relative to a stored equivalent seminorm.

Statement

Let X, Y be real normed spaces and let p be the seminorm of an equivalent-seminorm datum on X , with stored constants $c, C > 0$ satisfying $c\|x\| \leq p(x) \leq C\|x\|$ for every $x \in X$ and with p continuous. A continuous real-linear map $A : X \rightarrow Y$ is a contraction relative to this datum exactly when $\|Ax\| \leq p(x)$ for every $x \in X$.

Definition

For the given continuous linear map A , define

$$\text{IsContraction}_p(A) \iff \forall x \in X, \|Ax\| \leq p(x).$$

Here p is evaluated on the domain vector, and the norm is the codomain norm.

Assumptions

The map A is already continuous and real-linear. The domain carries its reference norm and an equivalent-seminorm datum p with positive comparison constants and continuity. The codomain Y is any real normed space, with the norm appearing on the left of the inequality. No dimension, completeness or bijectivity condition is imposed.

Conclusion

The declaration defines a proposition about the given map A . The separately defined contraction set is precisely $\{A : X \rightarrow Y \text{ continuous and real-linear} : \forall x, \|Ax\| \leq p(x)\}$.

When a later application takes $Y = \mathbb{R}^N$ in the form of real functions on a finite index set, its default norm is the supremum norm. This is a specialization of the same predicate. The definition itself supplies neither an operator nor an inverse.

Main citations

- Exact declaration and its source — `MathlibAnnex.EquivalentSeminorm.IsContraction`
- The equivalent-seminorm data — `MathlibAnnex.EquivalentSeminorm`
- The set of maps satisfying the pointwise bound — `MathlibAnnex.EquivalentSeminorm.contractionSet`

Lean source signature (exact)

```
def IsContraction (A : E →L[ℝ] F) : Prop := ∀ x, ‖A x‖ ≤ M.p x
```

In the source	Mathematical meaning
<pre>{E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] [NormedAddCommGroup F] [NormedSpace ℝ F]</pre>	The domain X and codomain Y are real normed spaces.
<pre>(M : EquivalentSeminorm E)</pre>	The supplied datum stores p, c, C with $c, C > 0$, $c\ x\ \leq p(x) \leq C\ x\ $ for every $x \in X$, and continuity of p .
<pre>(A : E →L[ℝ] F)</pre>	The given continuous real-linear map $A : X \rightarrow Y$. Continuity is included in this map type.

In the source	Mathematical meaning
<pre>IsContraction (A : E →L[ℝ] F) : Prop</pre>	A proposition about this given A , rather than a construction or existence statement.
<pre>∀ x, ‖A x‖ ≤ M.p x</pre>	The full defining right-hand side: for every $x \in X$, $\ Ax\ _Y \leq p(x)$. Here $M.p$ is the stored seminorm p .

▼ Exact surrounding binder context (separate excerpt)

```
namespace EquivalentSeminorm

variable {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
[NormedAddCommGroup F] [NormedSpace ℝ F] (M : EquivalentSeminorm E)
```

▼ Exact content identity

Declaration: MathlibAnnex.EquivalentSeminorm.IsContraction

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